CSE 242 Assignment 1

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Part 1

Q1. Matrix operations

Let $A = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$.

$$AB = \begin{pmatrix} 3 \cdot 1 + 4 \cdot 0 & 3 \cdot 2 + 4 \cdot (-1) \\ 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 \cdot 3 + 2 \cdot 2 & 1 \cdot 4 + 2 \cdot 1 \\ 0 \cdot 3 + (-1) \cdot 2 & 0 \cdot 4 + (-1) \cdot 1 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -2 & -1 \end{pmatrix}$$

$$(A + B)^\top = \begin{pmatrix} 4 & 6 \\ 2 & 0 \end{pmatrix}^\top = \begin{pmatrix} 4 & 2 \\ 6 & 0 \end{pmatrix}$$

$$AB^\top = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 + 4 \cdot 2 & 3 \cdot 0 + 4 \cdot (-1) \\ 2 \cdot 1 + 1 \cdot 2 & 2 \cdot 0 + 1 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 11 & -4 \\ 4 & -1 \end{pmatrix}$$

$AB \neq BA$, so matrix multiplication is not commutative. The traces are $\operatorname{tr}(A) = 3 + 1 = 4$ and $\operatorname{tr}(AB) = 3 + 3 = 6$. As a check, $\operatorname{tr}(BA) = 7 - 1 = 6 = \operatorname{tr}(AB)$, as Q3 predicts.

Answer. $AB = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$, $BA = \begin{pmatrix} 7 & 6 \\ -2 & -1 \end{pmatrix}$, $(A + B)^\top = \begin{pmatrix} 4 & 2 \\ 6 & 0 \end{pmatrix}$, $AB^\top = \begin{pmatrix} 11 & -4 \\ 4 & -1 \end{pmatrix}$, $\operatorname{tr}(A) = 4$, $\operatorname{tr}(AB) = 6$.

Q2. Eigenvalues and eigenvectors

Let $M = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. The characteristic polynomial is

$$\det(M - \lambda I) = (2 - \lambda)^2 - (-1)(-1) = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

For $\lambda = 1$, $M - I = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. The first row gives $v_1 - v_2 = 0$, so $v = (1, 1)^\top$.

For $\lambda = 3$, $M - 3I = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}$. The first row gives $-v_1 - v_2 = 0$, so $v = (1, -1)^\top$.

Verification of $Mv = \lambda v$:

$$M \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 - 1 \\ -1 + 2 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad M \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 + 1 \\ -1 - 2 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The eigenvectors are orthogonal, as expected for a symmetric matrix. The code uses `np.linalg.eigh`, which is designed for symmetric matrices. It returns real eigenvalues and an orthonormal set of eigenvectors, so a repeated eigenvalue still receives a full basis of its eigenspace. It returns unit vectors, so it reports $(1, 1)^\top/\sqrt{2}$ and $(1, -1)^\top/\sqrt{2}$ up to sign.

Answer. Characteristic polynomial $\lambda^2 - 4\lambda + 3$. Eigenpairs $\lambda = 1$, $v = (1, 1)^\top$ and $\lambda = 3$, $v = (1, -1)^\top$.

Q3. $\mathrm{tr}(AB) = \mathrm{tr}(BA)$

Let A be $m \times n$ and B be $n \times m$. The matrices need not be square. AB is $m \times m$ and BA is $n \times n$, so both traces are defined. By the definition of the matrix product, $(AB)_{ii} = \sum_{k=1}^{n} A_{ik} B_{ki}$, so

$$\mathrm{tr}(AB) = \sum_{i=1}^{m} \sum_{k=1}^{n} A_{ik} B_{ki}.$$

The sums are finite, so their order can be swapped. Scalar multiplication commutes, so $A_{ik} B_{ki} = B_{ki} A_{ik}$. Therefore

$$\mathrm{tr}(AB) = \sum_{k=1}^{n} \sum_{i=1}^{m} B_{ki} A_{ik} = \sum_{k=1}^{n} (BA)_{kk} = \mathrm{tr}(BA).$$

Answer. $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for all $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times m}$. Q1 confirms it with $\mathrm{tr}(AB) = \mathrm{tr}(BA) = 6$.

Q4. Inverse of C

Let $C = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$. Its determinant is $\det(C) = 4 \cdot 6 - 7 \cdot 2 = 24 - 14 = 10 \neq 0$. A square matrix is invertible if and only if its determinant is nonzero, so C is invertible. The 2×2 adjugate formula gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \Rightarrow \quad C^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}.$$

Verification:

$$CC^{-1} = \begin{pmatrix} 4(0.6) + 7(-0.2) & 4(-0.7) + 7(0.4) \\ 2(0.6) + 6(-0.2) & 2(-0.7) + 6(0.4) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The code uses `np.linalg.inv` on a float copy because the adjugate formula above covers only the 2×2 case and the function must handle $n \times n$ input.

Answer. $\det(C) = 10 \neq 0$, so C is invertible, and $C^{-1} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}$.

Part 2

Q1. Normalization, expectation and variance

$P(X = k) = c\,r^k/k!$ for $k = 0, 1, 2, \ldots$ The derivation uses a general rate $r > 0$ and sets $r = 3$ at the end.

Normalization. The Taylor series of the exponential is $e^r = \sum_{k=0}^{\infty} r^k/k!$. The probabilities must sum to 1, so

$$1 = \sum_{k=0}^{\infty} c\,\frac{r^k}{k!} = c\,e^r \quad \Rightarrow \quad c = e^{-r}.$$

Expectation. The $k = 0$ term is zero, and $k/k! = 1/(k-1)!$ for $k \geq 1$. Shifting the index with $j = k - 1$,

$$\mathbb{E}[X] = e^{-r} \sum_{k=1}^{\infty} \frac{r^k}{(k-1)!} = e^{-r}\,r \sum_{j=0}^{\infty} \frac{r^j}{j!} = e^{-r}\,r\,e^r = r.$$

Variance. The $k = 0$ and $k = 1$ terms of $\mathbb{E}[X(X-1)]$ are zero, and $k(k-1)/k! = 1/(k-2)!$ for $k \geq 2$. Shifting the index with $j = k - 2$,

$$\mathbb{E}[X(X-1)] = e^{-r} \sum_{k=2}^{\infty} \frac{r^k}{(k-2)!} = e^{-r}\,r^2 \sum_{j=0}^{\infty} \frac{r^j}{j!} = r^2.$$

Then $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = r^2 + r$, and by Q6,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = r^2 + r - r^2 = r.$$

This is the Poisson distribution, and the results agree with its known mean and variance.

Answer. For rate 3: $c = e^{-3} \approx 0.0498$, $\mathbb{E}[X] = 3$, $\mathrm{Var}(X) = 3$.

Q2. Event probabilities

Given $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \mid B) = 0.6$.

- Intersection, from the definition of conditional probability: $P(A \cap B) = P(A \mid B)\,P(B) = 0.6 \cdot 0.3 = 0.18$.
- Conditional, by Bayes' rule: $P(B \mid A) = \dfrac{P(A \mid B)\,P(B)}{P(A)} = \dfrac{0.18}{0.5} = 0.36$.
- Union, by inclusion-exclusion, since $P(A) + P(B)$ counts the intersection twice: $P(A \cup B) = 0.5 + 0.3 - 0.18 = 0.62$.

Answer. $P(A \cap B) = 0.18$, $P(B \mid A) = 0.36$, $P(A \cup B) = 0.62$.

Q3. Coin MLE and MAP

Let $\theta = P(\text{heads})$, h the number of heads and n the number of tosses. The data H, H, T, H, T gives $h = 3$, $n = 5$. The tosses are independent, so

$$L(\theta) = \theta^h (1-\theta)^{n-h}, \qquad \theta \in [0, 1].$$

MLE, interior case $0 < h < n$.

$$\log L(\theta) = h \log \theta + (n-h)\log(1-\theta), \qquad \frac{d}{d\theta}\log L = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \;\Rightarrow\; \hat{\theta}_{\text{MLE}} = \frac{h}{n}.$$

The second derivative $-h/\theta^2 - (n-h)/(1-\theta)^2$ is negative, so $\log L$ is concave and this critical point is the maximum. L is zero at both endpoints, so no boundary point beats it.

MLE, boundary cases. The derivative argument does not apply here because the maximum lies on the boundary.

- $h = 0$: $L(\theta) = (1-\theta)^n$ is strictly decreasing, so the maximum is at $\theta = 0$.
- $h = n$: $L(\theta) = \theta^n$ is strictly increasing, so the maximum is at $\theta = 1$.

Both agree with h/n, so the formula holds for all $0 \le h \le n$.

MAP. The prior $p(\theta) = 2\theta$ is a valid density on $[0,1]$ because $\int_0^1 2\theta\, d\theta = 1$. By Bayes' rule,

$$p(\theta \mid \text{data}) \propto \theta^h (1-\theta)^{n-h} \cdot 2\theta \propto \theta^{h+1}(1-\theta)^{n-h}.$$

The prior adds one to the exponent of θ, as if one extra head had been observed. For $h < n$,

$$\frac{d}{d\theta}\log p = \frac{h+1}{\theta} - \frac{n-h}{1-\theta} = 0 \;\Rightarrow\; \hat{\theta}_{\text{MAP}} = \frac{h+1}{n+1}.$$

The log posterior is concave and the posterior is zero at both endpoints, so this is the maximum. This includes $h = 0$, where the mode is $1/(n+1) > 0$ because the prior pulls the estimate away from 0. For $h = n$ the posterior θ^{n+1} is strictly increasing, so the maximum is at $\theta = 1$, which the formula also gives. Since $n \ge 1$, neither formula divides by zero.

Answer. $\hat{\theta}_{\text{MLE}} = 3/5 = 0.6$ and $\hat{\theta}_{\text{MAP}} = 4/6 = 2/3$. Boundary checks: $(h, n) = (0, 5)$ gives MLE 0 and MAP $1/6$. $(h, n) = (5, 5)$ gives MLE 1 and MAP 1.

Q4. Gaussian MLE

For i.i.d. samples $x_1, \ldots, x_n \sim \mathcal{N}(\mu, \sigma^2)$, the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Setting each partial derivative to zero:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0 \;\Rightarrow\; \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0 \;\Rightarrow\; \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2.$$

The variance MLE divides by n. It is the biased estimator. For the data $\{1, 3, 5, 7\}$ with $n = 4$:

- $\hat{\mu} = (1+3+5+7)/4 = 16/4 = 4$.
- The deviations are $-3, -1, 1, 3$, the squared deviations are $9, 1, 1, 9$, and their sum is 20.
- $\hat{\sigma}^2 = 20/4 = 5$. The unbiased $n-1$ divisor would give $20/3$, which is not the MLE.

Answer. $\hat{\mu}_{\text{MLE}} = 4$ and $\hat{\sigma}^2_{\text{MLE}} = 5$.

Q5. Gaussian MAP

One observation $x \sim \mathcal{N}(\mu, \sigma^2)$ with known variance σ^2, and prior $\mu \sim \mathcal{N}(\mu_0, \tau^2)$. Both σ^2 and τ^2 are variances. Given $x = 5$, $\sigma^2 = 4$, $\mu_0 = 0$, $\tau^2 = 1$. Dropping terms that do not depend on μ,

$$\log p(\mu \mid x) = -\frac{(x-\mu)^2}{2\sigma^2} - \frac{(\mu-\mu_0)^2}{2\tau^2} + \text{const.}$$

This is a quadratic in μ with negative leading coefficient, so the posterior is Gaussian and its mode is the unique critical point:

$$\frac{d}{d\mu}\log p(\mu \mid x) = \frac{x-\mu}{\sigma^2} - \frac{\mu-\mu_0}{\tau^2} = 0 \;\Rightarrow\; \hat{\mu}_{\text{MAP}} = \frac{x/\sigma^2 + \mu_0/\tau^2}{1/\sigma^2 + 1/\tau^2}.$$

This is the average of x and μ_0 weighted by their precisions. For a Gaussian the mode equals the mean, so it is also the posterior mean. With the given values,

$$\hat{\mu}_{\text{MAP}} = \frac{5/4 + 0/1}{1/4 + 1/1} = \frac{1.25}{1.25} = 1.$$

The prior has precision 1 and the observation has precision $1/4$, so the estimate is pulled strongly from $x = 5$ toward the prior mean 0.

Answer. $\hat{\mu}_{\text{MAP}} = 1$.

Q6. $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

Let $\mu = \mathbb{E}[X]$, a constant. Expanding the square and applying linearity of expectation,

$$\text{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

Answer. $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Part 3

Q1. Covariance and $\text{Var}(u^\top X) = u^\top \Sigma u$

For a random vector $X \in \mathbb{R}^d$ with mean $\mu = \mathbb{E}[X]$, the covariance matrix is

$$\Sigma = \mathbb{E}\big[(X-\mu)(X-\mu)^\top\big], \qquad \Sigma_{ij} = \text{Cov}(X_i, X_j).$$

Σ is symmetric because $(X-\mu)(X-\mu)^\top$ is symmetric. Let $u \in \mathbb{R}^d$ be fixed.

1. By linearity, $\mathbb{E}[u^\top X] = u^\top \mu$.
2. By the definition of variance, $\text{Var}(u^\top X) = \mathbb{E}\big[(u^\top X - u^\top \mu)^2\big] = \mathbb{E}\big[(u^\top (X-\mu))^2\big]$.
3. $u^\top (X-\mu)$ is a scalar, so it equals its transpose $(X-\mu)^\top u$. Hence $(u^\top (X-\mu))^2 = u^\top (X-\mu)(X-\mu)^\top u$.
4. u is not random, so linearity gives $\mathbb{E}\big[u^\top (X-\mu)(X-\mu)^\top u\big] = u^\top\, \mathbb{E}\big[(X-\mu)(X-\mu)^\top\big]\, u = u^\top \Sigma u$.

A variance is never negative, so $u^\top \Sigma u \geq 0$ for every u. Σ is therefore positive semidefinite and all its eigenvalues are nonnegative.

Answer. $\text{Var}(u^\top X) = u^\top \Sigma u$, and Σ is positive semidefinite.

Q2. Data generation and correlation

`generate_data` draws i.i.d. samples from $\mathcal{N}(m, S)$ by calling `multivariate_normal` on the generator `np.random.default_rng(seed)`. A fixed seed gives identical samples on every call. The defaults are

$$m = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 3 & 1.2 \\ 1.2 & 1 \end{pmatrix}.$$

The true correlation is $\rho = 1.2/\sqrt{3 \cdot 1} = 1.2/\sqrt{3} \approx 0.6928$. This is a clear positive correlation. When X_1 is above its mean, X_2 tends to be above its mean too, so the scatter forms an ellipse tilted upward. X_1 has variance 3 and X_2 has variance 1, so the cloud is also wider horizontally than vertically.

For the default sample (500 points, seed 0), the functions return

$$\hat{m} = \begin{pmatrix} 2.1524 \\ -0.9346 \end{pmatrix}, \qquad \hat{S} = \begin{pmatrix} 2.9162 & 1.1657 \\ 1.1657 & 0.9530 \end{pmatrix}, \qquad \hat{\rho} = 0.6992.$$

All three are close to the true values, as expected for 500 samples.

Answer. True correlation $\rho = 1.2/\sqrt{3} \approx 0.6928$. Sample correlation $\hat{\rho} = 0.6992$, a clear positive tilt.

Q3. PCA

PCA looks for the unit direction u along which the projection $u^\top X$ has the largest variance. By Q1 that variance is $u^\top \Sigma u$, so

$$u_1 = \arg\max_u \; u^\top \Sigma u \quad \text{subject to} \quad \|u\| = 1.$$

The constraint is needed because otherwise the objective grows without bound as u is scaled. The Lagrangian is $\mathcal{L}(u, \lambda) = u^\top \Sigma u - \lambda(u^\top u - 1)$. Σ is symmetric, so

$$\nabla_u \mathcal{L} = 2\Sigma u - 2\lambda u = 0 \;\Rightarrow\; \Sigma u = \lambda u.$$

Every critical point is an eigenvector of Σ, and there the objective equals $u^\top \Sigma u = \lambda u^\top u = \lambda$. The maximum is therefore the largest eigenvalue, attained at its eigenvector. The second component maximizes the same objective over unit vectors orthogonal to u_1, which gives the eigenvector of the second largest eigenvalue. Each eigenvalue is the variance of the data along its component.

Each step of `pca` implements part of this derivation:

(i) Centering subtracts the sample mean, producing the $X - \mu$ inside Σ.

(ii) The covariance is estimated as $\hat{\Sigma} = \frac{1}{n-1} X_c^\top X_c$, where X_c is the centered data. This is the sample version of the expectation with the unbiased divisor.

(iii) `np.linalg.eigh` solves $\Sigma u = \lambda u$, the Lagrangian condition. $\hat{\Sigma}$ is symmetric, so the eigenvectors are orthonormal. This holds when eigenvalues are tied (any orthonormal basis of the shared eigenspace is returned) and when $\hat{\Sigma}$ is rank-deficient (an eigenvalue is 0).

(iv) `eigh` returns ascending eigenvalues, so the order is reversed to put the largest variance first.

(v) By Q1 the eigenvalues are nonnegative. Tiny negative values from floating-point error are clipped to 0.

(vi) The scores $X_c U$ give $u_j^\top (x_i - \hat{\mu})$, the coordinates of each point in the new basis. The sample variance of column j equals λ_j.

For the default sample, the functions return (eigenvector signs are arbitrary)

$$\lambda_1 = 3.4585, \quad \lambda_2 = 0.4107, \qquad u_1 = \pm \begin{pmatrix} 0.9067 \\ 0.4218 \end{pmatrix}, \quad u_2 = \pm \begin{pmatrix} -0.4218 \\ 0.9067 \end{pmatrix}.$$

u_1 points 25.0° above the horizontal, along the long axis of the ellipse. The true covariance gives $\lambda_1 = 3.5620$, $\lambda_2 = 0.4380$ and an angle of 25.1°, so the sample estimate is close.

Answer. $\lambda_1 = 3.4585$ and $\lambda_2 = 0.4107$, with $u_1 = \pm(0.9067, 0.4218)^\top$ at 25.0° and $u_2 = \pm(-0.4218, 0.9067)^\top$ perpendicular to it.

Q4. Explained variance

The fraction of variance explained by component j is $\lambda_j / \sum_k \lambda_k$. The eigenvalues sum to $\mathrm{tr}(\hat{\Sigma})$, the total variance (here $3.8692 = 3.8692$), so the fractions are nonnegative and sum to 1.

For the default sample, PC1 explains 89.4% and PC2 explains 10.6%. The true covariance gives 89.1% and 10.9%.

Projecting onto PC1 alone reduces the data from 2D to 1D and keeps about 89.4% of the variance. The tradeoff:

- Benefits: each point is stored with half as many numbers, and later computations run on a smaller representation. If the variance along PC2 is mostly noise, discarding it also denoises the data.
- Cost: the 10.6% of variance along PC2 is lost. Points that differ only along PC2 become identical, and the reconstruction error equals the discarded variance. If that direction carries real signal, such as the feature separating two classes, the projection removes it.

PCA keeps the directions of largest variance, which are not always the directions most useful for a given task.

Answer. PC1 explains 89.4% and PC2 explains 10.6%. A 1D projection onto PC1 keeps most of the variance.

Figures

Figure 1 is produced by `make_plots` and saved by the main block. In the left panel, the samples form an ellipse tilted upward, which shows the positive correlation from Q2. Each arrow starts at the sample mean and has length $\sqrt{\lambda_j}$, the standard deviation along that component. The PC1 arrow ($\sqrt{\lambda_1} = 1.86$) runs along the long axis at about 25.0°. The PC2 arrow ($\sqrt{\lambda_2} = 0.64$) is perpendicular to it along the short axis. The axes use equal scaling, so the arrows appear perpendicular. The right panel shows the explained variance fractions from Q4.

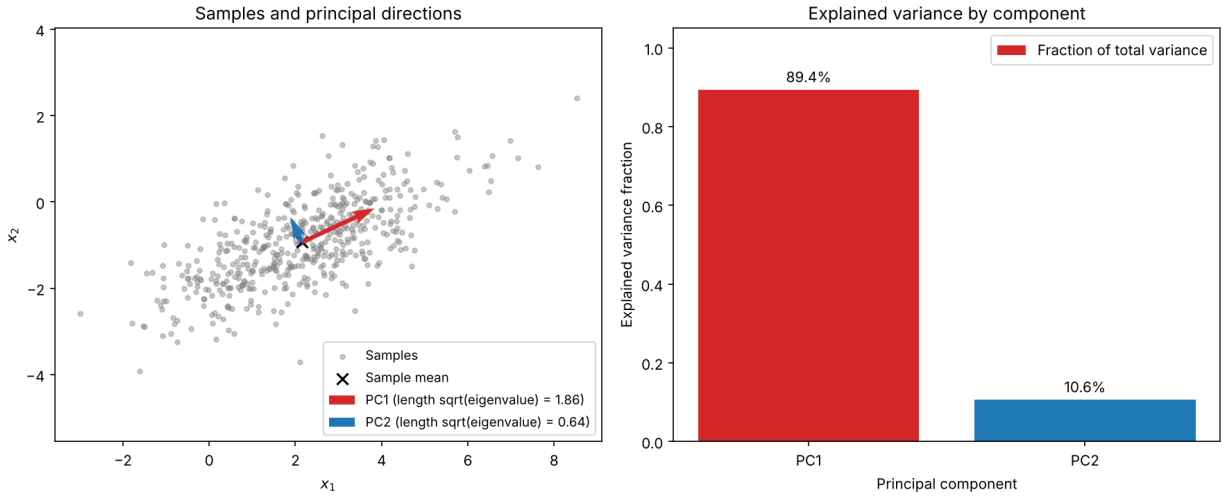


Figure 1: Left: samples with principal directions drawn from the sample mean, each scaled by $\sqrt{\lambda_j}$. Right: explained variance fraction of each component.

Reflection

Tasks completed. All parts. Part 1 (matrix operations, eigensystem, trace proof, matrix inverse), Part 2 (distribution moments, event probabilities, coin MLE and MAP, Gaussian MLE and MAP, variance identity) and Part 3 (covariance derivation, data generation, PCA, explained variance, figures).

Contributions. I worked on this assignment individually.

External resources. I used Claude (Anthropic's AI assistant) to help draft the written derivations and the code. I specified the approach for each question, checked the derivations, and tested every function on the default inputs and on other inputs, including boundary, tied and rank-deficient cases. I also used the NumPy and Matplotlib documentation.